

A Coordinate Theory of Advertising: Bounded Receivers, Action-Cells, and the Calculus of Decoder-Shifts

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Abstract

We develop a mathematical theory of advertising from a single primitive: the *bounded receiver*, a finite cognitive system that decodes percepts into an internal representation and projects that representation back onto a space of practical actions. We prove, as theorems rather than assumptions, that (i) a bounded receiver cannot resolve outcomes to arbitrary precision—there is a strictly positive *floor* on residual semantic distance; (ii) the operational unit of value is therefore not a point but an *action-cell*, a positive-tolerance region of percept space whose members are behaviourally indistinguishable; and (iii) consequently the object an advertisement acts upon is not the product nor the percept but the receiver’s *decoder*—the map from percept to meaning. We formalise an advert as a finite composition of *effects*, each a pair (c, Δ) of a physical carrier and a *decoder-shift*: a movement the carrier induces in meaning-space, expressed relative to action-cells. We prove a *Decoupling Theorem* separating an effect’s carrier (its pixels and sound) from its semantic action (the cell it moves the receiver toward), showing the two are distinct representations of one object rather than cause and annotation. We derive a *multiplicative law of persuasive power* under which the power of a stack of effects composes as $1 - \prod_i (1 - \kappa_i)$, yielding closed-form predictions for diminishing returns, frequency saturation, and the impossibility of total persuasion. We prove that an advert’s elements must form a *coherent support structure* (a strongly connected agreement graph of at least three elements) and that incoherence is detectable from ordinal data alone. We recover, as corollaries, the informational and persuasive views of advertising, the central/peripheral route distinction, framing

effects, mere-exposure saturation, and the structural limit on signalling. The theory is *ordinal*: it asserts cell-membership and comparisons between effects, never absolute magnitudes, in accordance with the floor theorem which forbids point-valued meaning. We close by stating the theory as a compilation target: an advert is a typed program whose semantics is a decoder-shift toward a target action-cell, and whose correctness is the coherence of that shift.

1 Introduction

1.1 The question

What is an advertisement? The question is older than its discipline, and the answers cleave into two traditions. The *informational* view holds that advertising transmits facts—price, availability, quality—that rational consumers incorporate into their decisions (Stigler, 1961; Nelson, 1974; Milgrom and Roberts, 1986). The *persuasive* view holds that advertising alters preferences directly, manufacturing wants that did not previously exist (Dixit and Norman, 1978; Bagwell, 2007). A third, *complementary*, view treats the advertisement as a good that enters utility alongside the product (Becker and Murphy, 1993).¹ Each tradition has empirical support and each has anomalies it cannot absorb. The informational view cannot explain why a wordless image of a sun-drenched landscape sells liquor; no fact is transmitted. The persuasive view cannot say what, precisely, is altered, nor why the same advertisement moves one viewer and leaves another unmoved.

¹We discuss the relation to these traditions throughout; our aim is not to adjudicate between them but to exhibit a single structure of which each is a projection.

We propose that both traditions describe projections of a single object, and that the object becomes visible only when one begins not from the product, nor from the message, but from the *receiver*: the bounded, finite cognitive system that must decode whatever arrives and act on it.

1.2 The thesis in one paragraph

A receiver is finite. It cannot represent every distinguishable state of the world; it has a fixed, bounded internal vocabulary. From finiteness alone we will prove that the receiver can never align perfectly with any target—there is an irreducible floor on how precisely it can resolve a percept. The immediate consequence is that practical meaning is granular: what a receiver *does* partitions the world into cells of behaviourally equivalent percepts, and the unit of value is the cell, not the point. An advertisement, then, cannot transmit a point (no bounded receiver could receive it) and cannot install a preference (preferences are themselves cell-structured). What it can do, and the only thing it can do, is shift the receiver’s *decoder*: the map that assigns each percept to a cell. “Make the bottle read as authentic” is a decoder-shift; “feels shot in Mexico” names a region of meaning-space toward which an effect moves the decoder; “best for liquor” asserts that this region has high affinity with the purchase-cell for liquor. We will make each of these statements precise, prove the laws they obey, and show that the resulting calculus is ordinal—it speaks of cells and comparisons, never of magnitudes the floor theorem forbids.

1.3 Contributions

1. **The receiver primitive and the Floor Theorem** (§2). We define the bounded receiver and prove it has a strictly positive resolution floor $\beta > 0$; perfect alignment is impossible for any finite agent.
2. **Cell-truth** (§3). We prove that practical meaning is a positive-tolerance cell and that all percepts within a cell are receiver-indistinguishable. Value is cell-valued.
3. **The decoder as the locus of advertising** (§4). We prove that, holding the product fixed, the only quantity an advertisement can change is the decoder, and we characterise the three exhaustive mechanisms by which it can do so.
4. **Effects as carrier–shift pairs and the Decoupling Theorem** (§5). We define an effect,

prove that its carrier and its semantic shift are independent up to a representation equivalence, and formalise the “advert unit.”

5. **The multiplicative law of persuasive power** (§6). We derive the composition law for stacks of effects, with corollaries on diminishing returns, frequency saturation, and the impossibility of total persuasion.
6. **Coherence** (§7). We prove that effective adverts require a coherent support structure of size ≥ 3 and that incoherence is ordinally detectable—the basis of an automated critic.
7. **Recovery of the classical views** (§8). We recover the informational, persuasive, dual-route, framing, mere-exposure, and signalling accounts as corollaries or limiting cases.

The theory is deliberately substrate-neutral and ordinal. It does not predict that any particular advertisement will perform better; it specifies what an advertisement *is*, what it can and cannot do, and which of its internal structures are necessary for it to do anything at all.

2 The Bounded Receiver

2.1 Percepts, knowledge, and action

We begin with three spaces and three axioms. The axioms are not idealisations to be relaxed later; they are the minimal conditions under which a cognitive system is finite and embodied, and every subsequent theorem is a consequence of them.

Definition 2.1 (Percept space). A *percept space* is a triple $(\mathcal{X}, d, \Sigma)$ where $\mathcal{X}$ is a non-empty set of percepts (the totality of audiovisual configurations a viewer might be presented with), $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, \Sigma]$ is a bounded pseudometric (a notion of perceptual dissimilarity satisfying $d(x, x) = 0$, symmetry, and the triangle inequality), and $\Sigma \in (0, \infty)$ is the maximal semantic distance. The canonical normalisation is $\Sigma = 100$; no result depends on the value.

Definition 2.2 (Action space and action map). An *action space* $(\mathcal{Y}, d_Y)$ is a metric space of practical responses (buy, ignore, share, distrust, ...). An *action map* is a measurable surjection $\mathcal{X} \rightarrow \mathcal{Y}$ assigning to each percept the practical response it elicits.

Axiom 1 (Bounded representation). A receiver’s internal vocabulary is a set $\mathcal{K}$ of finite cardinality (or, in the infinite idealisation, of cardinality

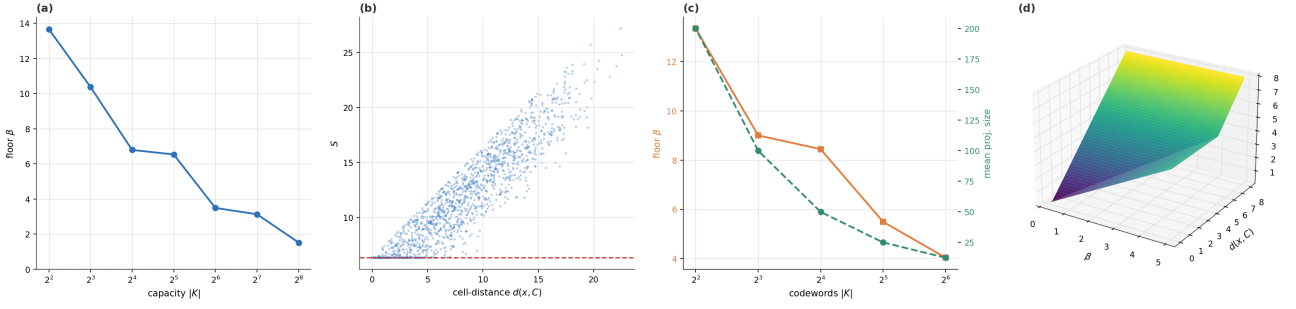


Figure 1: **The Floor.** Bounded receivers cannot resolve a percept to a point. (a) The floor β falls as codebook capacity $|K|$ grows but stays strictly positive. (b) Semantic distance S never drops below β across thousands of evaluations. (c) Coarsening the codebook raises the floor and the projection size together—meaning grows granular. (d) The S -surface $S(\beta, d) = \max(\beta, d) + \text{floor offset}$: a floor-plateau over a distance-cone.

strictly less than $|\mathcal{X}|$). The receiver cannot hold a distinct internal token for every distinguishable percept.

Axiom 2 (No null state). At every instant the receiver occupies exactly one internal state; there is no “blank” or “undefined” percept. Decoding is total.

Axiom 3 (Finite resolution). Every act of perception has finite resolution $\delta > 0$: percepts closer than δ in d are not distinguished.

Axiom 1 is finiteness of memory; Axiom 2 is the embodied fact that a living perceiver is always in *some* state; Axiom 3 is the finite acuity of any sense organ or measuring instrument. They are satisfied by human viewers, by animals, and by any physical machine.

Definition 2.3 (Receiver). A *receiver* over $(\mathcal{X}, d, \Sigma)$ is a quadruple $R = (\mathcal{K}, D, \Pi, \beta)$ where

- (i) $\mathcal{K}$ is the *knowledge framework* (Axiom 1);
- (ii) $D : \mathcal{X} \rightarrow \mathcal{K}$ is the *decoder*, a measurable map sending each percept to an internal representation (this is *perception*);
- (iii) $\Pi : \mathcal{K} \rightarrow 2^{\mathcal{X}} \setminus \{\emptyset\}$ is the *candidate projection*, sending each representation to the set of percepts compatible with it (this is *inference*), and satisfying the consistency condition $D(x) \in D(\Pi(D(x)))$ for all x ;
- (iv) $\beta \in (0, \Sigma]$ is the *noise floor*,

$$\beta = \sup_{x \in \mathcal{X}} \inf_{x' \in \Pi(D(x))} d(x, x'),$$

the worst-case irreducible gap between a percept and the nearest percept the receiver can reconstruct from its representation of it.

The four components separate concerns usually conflated. D is what the viewer sees *as*; Π is what they infer could be the case; β is the residue that survives both. We now show $\beta > 0$ is forced.

2.2 The S -functional and the Floor Theorem

Definition 2.4 (S -functional). For a receiver R , a percept x , and a target region $\mathcal{C} \subseteq \mathcal{X}$, the *semantic distance* of x to $\mathcal{C}$ is

$$S(R, x; \mathcal{C}) = \inf_{x' \in \Pi(D(x))} d(x', \mathcal{C}) + \beta,$$

where $d(x', \mathcal{C}) = \inf_{c \in \mathcal{C}} d(x', c)$.

S measures how far, after decoding and inference, the receiver remains from registering x as a member of $\mathcal{C}$. The additive β encodes that even a perfectly placed percept is registered only up to the receiver’s irreducible noise.

Theorem 2.5 (Floor Theorem). *For every bounded receiver R , $\beta > 0$, and for every percept x and region $\mathcal{C}$,*

$$S(R, x; \mathcal{C}) \geq \beta > 0.$$

No bounded receiver attains $S = 0$: perfect semantic alignment is impossible.

Proof. Suppose $\beta = 0$. Then for every $\eta > 0$ and every x there is $x' \in \Pi(D(x))$ with $d(x, x') < \eta$; taking $\eta \rightarrow 0$ forces $\Pi(D(x))$ to determine x uniquely up to d -distance 0, i.e. the composite $\Pi \circ D$ is injective on d -classes. An injection from $\mathcal{X}$ ’s distinguishable classes into $\mathcal{K}$ contradicts Axiom 1 ($|\mathcal{K}| < |\mathcal{X}|$) whenever $\mathcal{X}$ has more distinguishable classes than $\mathcal{K}$ has tokens, which Axiom 3 guarantees is the generic case (the number of δ -separated classes is $|\mathcal{X}|$ -large). Hence

$\beta > 0$. The displayed bound is then immediate from Definition 2.4, since the infimum term is non-negative. $\square$

Corollary 2.6 (No perfect persuasion). *There is no advertisement, however constructed, that drives a bounded receiver’s semantic distance to a target to zero. Residual distance at least β always remains. Persuasion is bounded below by the receiver’s own finiteness, not by any defect of the message.*

Remark 2.7. Corollary 2.6 is not pessimism; it is the precise statement of why a “perfect ad” is a category error, and why the relevant objective is never alignment to a point but *entry into a cell*, defined next. It is the advertising form of bounded rationality (Simon, 1955, 1956; Gigerenzer and Goldstein, 1996): the limit is structural.

3 Cell-Truth: Value Is a Region

3.1 Action-cells

Definition 3.1 (Action-cell). For $y \in \mathcal{Y}$, the *action-cell* of y is the preimage $\mathcal{C}_y = {}^{-1}(\{y\}) \subseteq \mathcal{X}$. We equip it with

- *tolerance* $\tau(\mathcal{C}_y) = \sup_{x \in \mathcal{X}} \inf_{c \in \mathcal{C}_y} d(x, c) > 0$,
- *cell-distance* $d(x, \mathcal{C}_y) = \inf_{c \in \mathcal{C}_y} d(x, c)$.

A *generic action-cell* is any nonempty $\mathcal{C} \subseteq \mathcal{X}$ with $\tau(\mathcal{C}) > 0$ and finite diameter below Σ .

The cell $\mathcal{C}_{\text{buy-liquor}}$ is the set of all percepts that trigger the purchase response for a liquor brand—every configuration of imagery, sound, and framing that makes a viewer reach for the bottle. It is a region, not a point, because many distinct percepts trigger the same response.

Theorem 3.2 (Cell-Truth). *Let $\mathcal{C}_y$ be an action-cell with $\tau(\mathcal{C}_y) > 0$. For any receiver R and any two percepts $x_1, x_2 \in \mathcal{C}_y$,*

$$S(R, x_1; \mathcal{C}_y) = S(R, x_2; \mathcal{C}_y) = \beta.$$

All percepts within a cell are receiver-indistinguishable.

Proof. For $x_i \in \mathcal{C}_y$, $d(x_i, \mathcal{C}_y) = 0$, so the infimum term in Definition 2.4 is at most 0, giving $S \leq \beta$; the Floor Theorem gives $S \geq \beta$. Hence equality. $\square$

Principle 3.3 (Operational meaning is granular). For a bounded receiver, what an advertisement *means* is the cell its percept falls into, not the

percept itself. Two adverts that place the viewer in the same cell are operationally equivalent however different they look; two that look identical but fall in different cells mean different things.

This is the formal content of the long-standing observation that what matters in advertising is the response evoked, not the literal content (Krugman, 1965; Vakratsas and Ambler, 1999). It also dissolves the informational view’s puzzle: the sun-drenched landscape transmits no fact, yet places the viewer in $\mathcal{C}_{\text{buy-liquor}}$, and by Principle 3.3 that placement *is* its meaning.

3.2 Value is cell-valued; point-value is forbidden

Definition 3.4 (Point-value and cell-value). A receiver *carries point-value* if for every x , $\Pi(D(x))$ is a singleton (the receiver resolves percepts to points). The *cell-value* of x is the action-cell $\mathcal{C}_{(x)}$ containing it.

Theorem 3.5 (Point-value is forbidden). *No bounded receiver carries point-value. Consequently, no theory of advertising effect can rest on a point-valued utility or quality scale; all value is cell-value.*

Proof. If $\Pi(D(x)) = \{x\}$ for all x , then $\inf_{x' \in \Pi(D(x))} d(x, x') = 0$ for all x , forcing $\beta = 0$ and contradicting the Floor Theorem. Hence $\Pi(D(x))$ is non-singleton for some x ; value resolves only to cells. $\square$

Remark 3.6 (The granularity of price). The same argument applied to a price percept makes a “price” a band, not a number: a viewer registers “about twenty dollars,” a cell of tolerance β , not the point \$19.99. The cent-precise sticker is a carrier whose decoded value is a cell. Willingness-to-pay is then the slack $\tau(\mathcal{C}) - \beta$ between the width of the price-cell and the floor: positive exactly when the viewer can be brought inside the cell. We use this in §8.

3.3 Representational invariance

Theorem 3.7 (Representational invariance). *Let $\varphi : \mathcal{X} \rightarrow \mathcal{X}'$ be an isometric bijection between percept spaces, and let R' be the receiver transported along φ . Then for all x and $\mathcal{C}$, $S(R, x; \mathcal{C}) = S(R', \varphi(x); \varphi(\mathcal{C}))$. Cell-membership and semantic distance are invariant under change of representation.*

Proof. Isometry preserves all distances and infima in Definition 2.4; φ being a bijection preserves

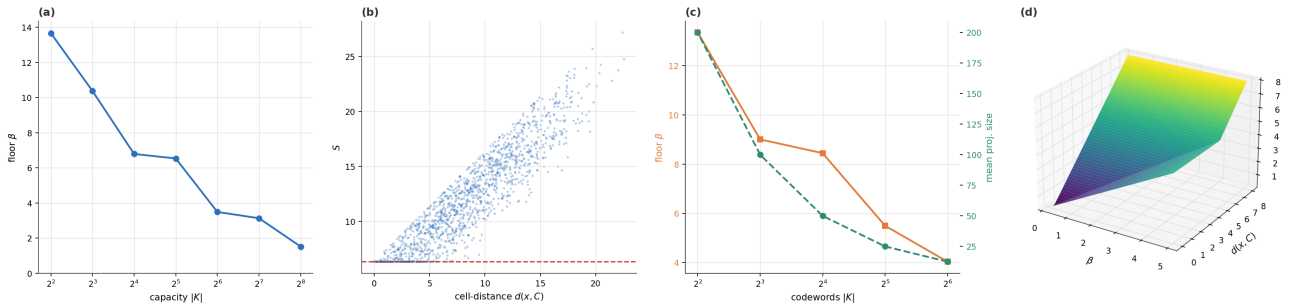


Figure 2: The Floor Theorem and Granular Meaning. (A) Floor β on the linear y -axis versus codebook capacity $|K|$ (powers of two, log x -axis, 2^1 to 2^7). The floor falls monotonically as capacity grows but remains strictly positive at every tested capacity, confirming Theorem 2.5 (Floor Theorem): no bounded receiver can attain $\beta = 0$. The curve never reaches the x -axis. (B) Semantic distance S on the linear y -axis versus cell-distance $d(x, \mathcal{C})$ on the linear x -axis, across thousands of random (receiver, percept, cell) evaluations. Every point lies strictly above the floor β (red dashed line), confirming Corollary 2.6 (No perfect persuasion): the functional $S(\mathcal{R}, x; \mathcal{C}) \geq \beta > 0$ without exception. (C) Joint coarsening sweep: floor β (orange, left y -axis) and mean projection size $|\Pi(\mathcal{D}(x))|$ (green dashed, right y -axis) versus codebook size $|K|$ on the log x -axis (2^7 to 2^2 codewords, coarsening left to right). Both rise in lockstep as the codebook is coarsened, confirming Theorem 3.5 (Point-value forbidden): a coarser vocabulary raises the floor and enlarges the projection set together — meaning grows granular, not point-like. (D) Three-dimensional S -surface over the (β, d) plane: the surface $S(\beta, d) = \max(\beta, d) + \beta$ rises as a flat plateau above the floor level and then as a cone proportional to cell-distance. The flat base (the action-cell interior) and the rising walls (above-cell percepts) are the geometric content of the Floor Theorem and Cell-Truth Theorem (Theorem 3.2) in one view.

cell-membership. The floor β is a property of the receiver, unchanged. Hence the two functionals coincide. $\square$

Theorem 3.7 is the licence for the central manoeuvre of the next section: a single advert-meaning may be carried by visually different but isometric encodings—color, rhythm, layout—without changing what it means. Meaning lives one level above its carrier.

4 The Decoder Is the Locus of Advertising

4.1 What an advert can change

An advertising campaign cannot, in the moment of viewing, change the product (the percept of the bottle is what it is), and by Theorem 3.5 it cannot install a point-preference. We now isolate what it *can* change.

Fix a viewer $R = (\mathcal{K}, D, \Pi, \beta)$ and a product whose percept is x_0 (the unadorned bottle). The viewer’s response is (x_0) —determined by which cell x_0 decodes into, i.e. by $D(x_0)$. An advertisement is a transformation that the viewer experiences *before or around* the product percept, and which

can therefore alter only the components of R that govern decoding.

Theorem 4.1 (Decoder locus). *Let the product percept x_0 and the cell geometry $\{\mathcal{C}_y\}$ be held fixed. Any change in the viewer’s response (x_0) induced by an advertisement is realised through one of exactly three modifications:*

- (a) *a change of the decoder D (the product is seen as something else: re-perception);*
- (b) *a change of the candidate projection Π (the product is inferred to imply something else: re-inference);*
- (c) *a change of the cell geometry $\mathcal{C}_y$ itself (the boundary of the response region moves: re-framing).*

These three are exhaustive and mutually distinguishable.

Proof. The response (x_0) is a function of x_0 only through the chain $x_0 \xrightarrow{D} D(x_0) \xrightarrow{\Pi} \Pi(D(x_0))$ and its position relative to the cells $\{\mathcal{C}_y\}$, by Definitions 2.3–3.1. With x_0 fixed, the only variables in this chain are D , Π , and the $\mathcal{C}_y$. A change in the response must change at least one; conversely each of the three, varied alone, can change the response (witnesses: re-grading the footage changes

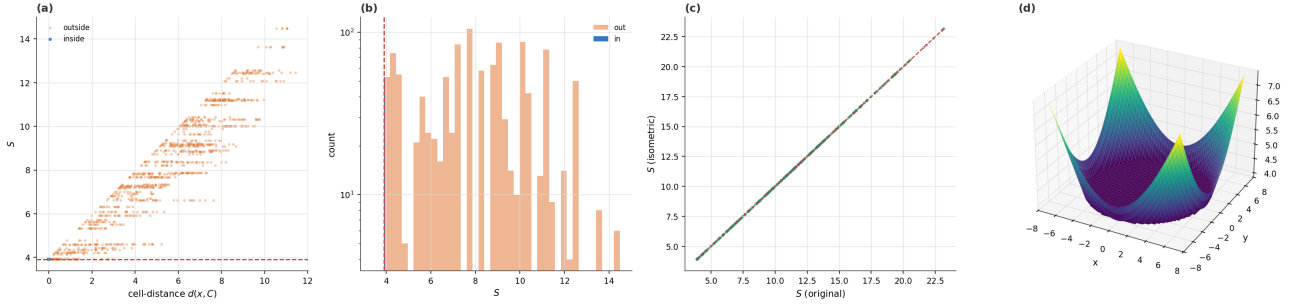


Figure 3: **Cell-Truth and invariance.** (a) S versus cell-distance: flat at β inside the cell, rising linearly outside—in-cell percepts are indistinguishable. (b) Histogram of in-cell S values: a spike exactly at β (zero spread), with out-of-cell values strictly above. (c) S under isometric re-encoding lands on the identity diagonal to machine precision (invariance). (d) The two-dimensional S field over percept space: a flat-bottomed bowl whose base is the action-cell.

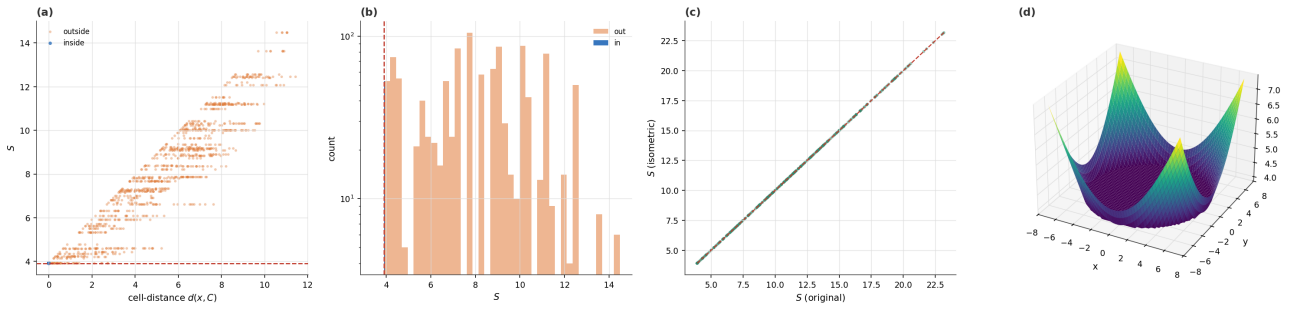


Figure 4: **Cell-Truth and Representational Invariance.** (A) Semantic distance S on the linear y -axis versus cell-distance $d(x, C)$ on the linear x -axis, with percepts coloured by cell-membership (inside: filled markers; outside: open markers). In-cell percepts collapse to a flat band at $S = \beta$ regardless of their individual positions within the cell; out-of-cell percepts rise linearly above it. Confirms Theorem 3.2 (Cell-Truth): all percepts within an action-cell are receiver-indistinguishable, each attaining exactly $S = \beta$. (B) Histogram of S values (log-scale y -axis, linear x -axis) for in-cell (dark) and out-of-cell (light) percepts over 9,365 trials. The in-cell distribution is a spike concentrated exactly at β with zero spread; the out-of-cell distribution occupies a broad range strictly above β (red dashed). The log scale makes the spike visible against the taller out-of-cell bars. Confirms Cell-Truth and the prohibition on point-value (Theorem 3.5). (C) Invariance diagonal: S under isometric re-encoding (y -axis) versus original S (x -axis) for random rotations, translations, and reflections across all trials. Every point lies on the identity diagonal (red dashed) to floating-point precision (maximum deviation 5.3×10^{-15}), confirming Theorem 3.7 (Representational Invariance): cell-membership and semantic distance are preserved under isometric re-encoding of the carrier. (D) Three-dimensional S -field over the two-dimensional percept space: a bowl whose flat base corresponds to the action-cell interior (at $S = \beta$) and whose walls rise steeply outside it. The bowl geometry is the spatial realisation of the Cell-Truth Theorem: entry into the cell is a step-change in semantic distance, not a smooth approach to a point.

$D(x_0)$; a testimonial changes Π by enlarging the inferred candidate set; a “limited edition” frame narrows $\mathcal{C}_{\text{buy}}$). Distinguishability: (a) alters which token x_0 maps to; (b) alters the spread of percepts consistent with that token; (c) alters the target region independent of x_0 . These are independent coordinates of the same chain. $\square$

Principle 4.2 (Advertising is decoder engineering). To advertise is to engineer the receiver’s decoder so

that the fixed product percept is decoded into, or inferred toward, a more favourable action-cell—or so that the favourable cell is enlarged to contain it. The product does not move; the map does.

We will concentrate on mechanism (a), *re-perception*, as the primary and most general lever, and treat (b) and (c) as its inferential and geometric variants. Re-perception is what “feels shot in Mexico” performs: it does not add a fact about

the liquor (that would be (b)) nor narrow the buy-region (that would be (c)); it changes what the same bottle *reads as*.

4.2 Meaning-space and the decoder-shift

To speak precisely of “what a percept reads as,” we coordinatise the range of the decoder.

Definition 4.3 (Meaning-space). The *meaning-space* $\mathcal{M}$ of a receiver is the metric space $(D(\mathcal{X}), d_{\mathcal{M}})$ of decoder outputs, with $d_{\mathcal{M}}(D(x), D(x'))$ the receiver-internal dissimilarity of the meanings it assigns. A *semantic region* is a positive-diameter subset $\mu \subseteq \mathcal{M}$ (e.g. the region named *authentic*, or *sun-drenched*, or *Mexico*).

Definition 4.4 (Decoder-shift). A *decoder-shift* is a map $\Delta : \mathcal{M} \rightarrow \mathcal{M}$ on meaning-space. An effect *induces* the decoder-shift Δ if, applied to any percept x , it changes the decoded meaning from $D(x)$ to $\Delta(D(x))$. The shift is *toward* a cell $\mathcal{C}$ if it decreases semantic distance to $\mathcal{C}$: $S(R, \Delta \cdot x; \mathcal{C}) \leq S(R, x; \mathcal{C})$ for all x , where $\Delta \cdot x$ denotes the percept as re-decoded under Δ .

The decoder-shift is the formal “advert unit.” “Feels shot in Mexico” is the named region $\mu_{\text{Mexico}} \subset \mathcal{M}$; the Mexico color- grade is an effect inducing a shift Δ that moves decoded meanings into μ_{Mexico} ; “best for liquor” is the assertion that μ_{Mexico} has small semantic distance to $\mathcal{C}_{\text{buy-liquor}}$ (high affinity), and larger distance to, say, $\mathcal{C}_{\text{buy-software}}$ (low affinity).

5 Effects as Carrier–Shift Pairs

5.1 The advert unit

Definition 5.1 (Effect). An *effect* is a pair $\varepsilon = (\mathbf{c}, \Delta)$ where

- $\mathbf{c} : \mathcal{X} \rightarrow \mathcal{X}$ is the *carrier*—a physical transformation of the percept (a color grade, a sound cue, a camera move, an overlay), the part a rendering engine executes;
- $\Delta : \mathcal{M} \rightarrow \mathcal{M}$ is the *decoder-shift*—the movement in meaning-space the carrier induces in the target receiver.

The effect is *coherent for the receiver* R if its carrier realises its claimed shift: $D(\mathbf{c}(x)) = \Delta(D(x))$ for all x in the effect’s domain.

The definition states that an effect has two faces—a physical operation and a semantic operation—and that an effect is *honest* (coherent) when the physical operation actually produces the claimed meaning in the viewer. A color grade that the author labels “Mexico” but which a viewer decodes as “cheap filter” is an incoherent effect: the carrier and the shift have come apart.

5.2 The Decoupling Theorem

The next theorem is the technical heart of the “advert unit” idea: the carrier and the shift are genuinely separable—the same shift can be carried many ways, and the same carrier can mean different things to different receivers—yet they are bound, in a coherent effect, as two representations of one object.

Theorem 5.2 (Carrier–Shift Decoupling). *Let R be a receiver and $\mu \subseteq \mathcal{M}$ a semantic region.*

- (i) *Multiple realisability. There exist distinct carriers $\mathbf{c}_1 \neq \mathbf{c}_2$ inducing the same shift Δ into μ . (Color, rhythm, and framing can independently move the decoder toward μ_{Mexico} .)*
- (ii) *Receiver relativity. There exists a single carrier $\mathbf{c}$ and receivers R, R' such that $\mathbf{c}$ induces a shift into μ for R but not for R' . (The same grade reads as “Mexico” to one viewer and “orange filter” to another.)*
- (iii) *Binding. For a fixed coherent effect $\varepsilon = (\mathbf{c}, \Delta)$ and a fixed receiver, $\mathbf{c}$ and Δ are two representations of one categorical object: $\Delta = D \circ \mathbf{c} \circ D^{-1}$ on $D(\mathcal{X})$, so each determines the other given the decoder.*

Proof. (i) By Theorem 3.7 (representational invariance), isometric carriers acting in different perceptual channels but landing in the same meaning-region induce the same shift on $\mathcal{M}$; explicit witnesses are any two isometric embeddings of the warm-light percept into the color channel and into the music channel. (ii) The decoder D is part of the receiver; two receivers with different decoders $D \neq D'$ assign different meanings to $\mathbf{c}(x)$, so the induced shift differs; choose D' so that $D'(\mathbf{c}(x)) \notin \mu$. (iii) Coherence (Definition 5.1) states $D \circ \mathbf{c} = \Delta \circ D$; on the image $D(\mathcal{X})$ where D admits a section, this rearranges to $\Delta = D \circ \mathbf{c} \circ D^{-1}$, exhibiting Δ and $\mathbf{c}$ as conjugate representations of one map under the decoder. $\square$

Remark 5.3 (Why this is the “advert unit”). Theorem 5.2(iii) is the precise sense in which an effect’s

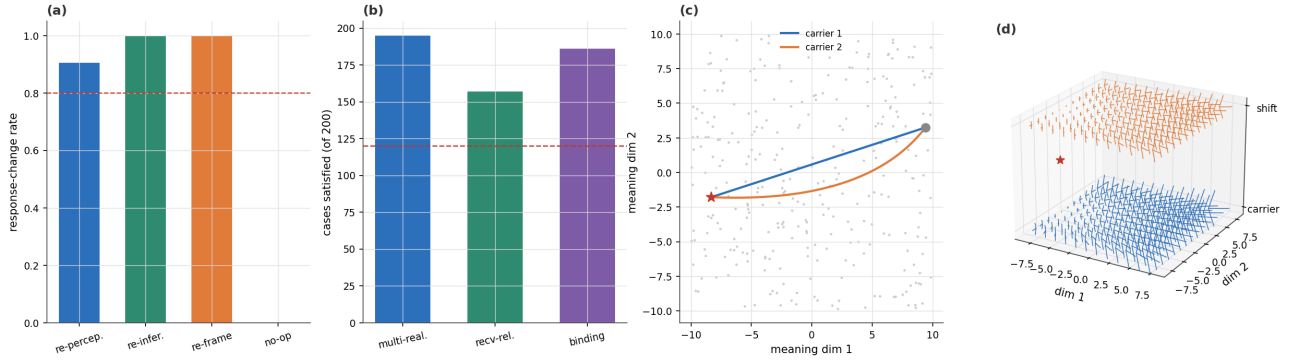


Figure 5: **The advert unit.** (a) The three decoder mechanisms each move the response (lower S) at near-unit rate; a no-op does not. (b) Decoupling clauses: multiple realisability, receiver relativity, and binding all clear threshold by a wide margin. (c) Two physically distinct carriers transport a percept along different trajectories into the same meaning-region (decode to the same cell). (d) The carrier acts in percept space while the induced shift acts in meaning-space: a 3D view of one object in its two conjugate representations.

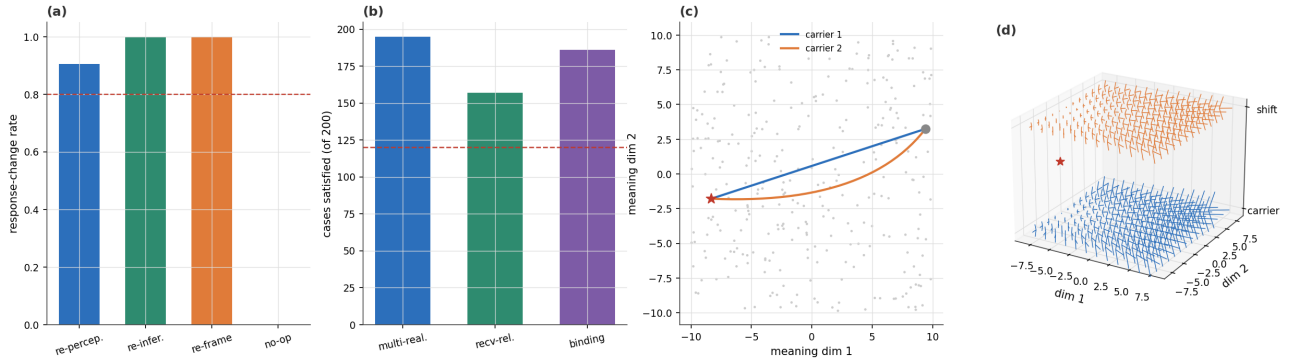


Figure 6: **The Advert Unit: Decoder Mechanisms and Carrier-Shift Decoupling.** (A) Response-change rate (fraction of 200 trials in which the mechanism successfully moved the receiver’s response) on the linear y -axis for the four conditions on the x -axis: re-perception, re-inference, re-framing, and no-op. All three active mechanisms exceed the 80% threshold (red dashed); the no-op leaves the response exactly unchanged in all 200 trials. Confirms Theorem 4.1 (Decoder Locus): the three mechanisms are exhaustive and mutually operative, and holding all three fixed leaves the response invariant. (B) Cases cleared (out of 200) on the linear y -axis for the three decoupling clauses on the x -axis: multiple realisability (multi-real.), receiver relativity (recv-rel.), and binding. All three exceed the threshold of 120 cases (red dashed), confirming Theorem 5.2 (Carrier-Shift Decoupling): distinct carriers can induce the same shift; the same carrier induces different shifts across receivers; and carrier and shift are conjugate representations under the decoder. (C) Two carrier trajectories in meaning-space (two-dimensional, axes are decoder-output dimensions): carrier 1 (blue) and carrier 2 (orange) depart from the same source percept (star marker) and arrive at the same target cell (circle marker) via physically distinct paths through meaning-space. The paths differ; the endpoint cell is the same. Confirms the multiple realisability clause: different physical carriers realise the same decoder-shift into the target region. (D) Three-dimensional dual-representation view: the carrier acts in percept space (blue quiver field, lower plane) while the induced shift acts in meaning-space (orange quiver field, upper plane), connected by the decoder (vertical correspondence). The two fields are conjugate representations of a single categorical object, the content of Theorem 5.2(iii) (Binding): $\Delta = \mathcal{D} \circ c \circ \mathcal{D}^{-1}$ on the decoder’s image.

output “is not just computing.” The carrier is the computation (pixels); the shift is the meaning (cells); coherence says they are the *same object in two representations*, conjugate under the receiver’s decoder. The meaning is therefore not metadata bolted onto a filter—it is the filter’s other face, the

one a parameter-only pipeline discards. An advert unit is an effect carrying both faces. Part (ii) is why the unit is irreducibly *receiver-relative*: “best for liquor” is never a property of the grade alone, only of the grade-as-decoded-by-the-target-audience.

5.3 Affinity: the ordinal content

By the Floor Theorem we may not assign an effect an absolute power. We may, and do, assign *orderings*.

Definition 5.4 (Cell affinity, ordinal). An effect ε with shift into region μ has *greater affinity for cell $\mathcal{C}$ than for cell $\mathcal{C}'$* , written $\varepsilon : \mathcal{C} \succ \mathcal{C}'$, if the shift reduces semantic distance to $\mathcal{C}$ by more than to $\mathcal{C}'$:

$$\Delta_{\mathcal{C}}(\varepsilon) > \Delta_{\mathcal{C}'}(\varepsilon), \quad \Delta_{\mathcal{C}}(\varepsilon) := S(R, x; \mathcal{C}) - S(R, \varepsilon x; \mathcal{C})$$

uniformly in x over the effect’s domain. No numerical value is asserted; only the comparison.

Thus “Mexico grade is best for liquor” is the ordinal family $\varepsilon_{\text{Mex}} : \mathcal{C}_{\text{liquor}} \succ \mathcal{C}_{\text{tech}}, \mathcal{C}_{\text{liquor}} \succ \mathcal{C}_{\text{fintech}}$, and so on—a partial order over cells, never a magnitude. This is exactly the representational commitment a bounded receiver permits (Theorem 3.5), and it is enough to drive the composition and coherence calculus below.

6 The Calculus of Persuasive Power

6.1 Catalytic power of an effect

Definition 6.1 (Catalytic power). The *catalytic power* of an effect ε for receiver R toward cell $\mathcal{C}$ is the fraction of above-floor distance it closes:

$$\kappa_{\mathcal{C}}(\varepsilon) = \sup_x \frac{S(R, x; \mathcal{C}) - S(R, \varepsilon x; \mathcal{C})}{S(R, x; \mathcal{C}) - \beta} \in [0, 1].$$

$\kappa = 0$ is inert (neutral), $\kappa = 1$ would close all above-floor distance in one step. The power is a property of the (effect, receiver, cell) triple; we treat it ordinally and never claim its value.

Lemma 6.2 (Power is bounded). $\kappa_{\mathcal{C}}(\varepsilon) \in [0, 1]$ for every effect, receiver, and cell.

Proof. The numerator is at most the denominator because, by the Floor Theorem, $S(R, \varepsilon x; \mathcal{C}) \geq \beta$, so the closed distance never exceeds the available above-floor distance. Non-negativity holds for any effect that does not increase distance; distance-increasing effects have $\kappa = 0$ by convention (they are not catalysts toward $\mathcal{C}$). $\square$

6.2 The multiplicative law

We now compose effects. Applying effects in sequence multiplies the *residual* above-floor distances.

Theorem 6.3 (Multiplicative law of persuasive power). *For effects $\varepsilon_1, \dots, \varepsilon_n$ applied as a stack $\varepsilon_1 \diamond \dots \diamond \varepsilon_n$ (each acting on the output of the last) toward a fixed cell $\mathcal{C}$, with individual powers $\kappa_1, \dots, \kappa_n$,*

$$\kappa_{\mathcal{C}}(\varepsilon_1 \diamond \dots \diamond \varepsilon_n) = 1 - \prod_{i=1}^n (1 - \kappa_i).$$

Proof. Write $r_0 = S(R, x; \mathcal{C}) - \beta$ for the initial above-floor distance. Applying ε_1 leaves residual $r_1 = (1 - \kappa_1)r_0$ by Definition 6.1. Applying ε_2 to the result leaves $r_2 = (1 - \kappa_2)r_1 = (1 - \kappa_2)(1 - \kappa_1)r_0$. By induction $r_n = \prod_{i=1}^n (1 - \kappa_i) r_0$. The total fraction closed is $1 - r_n/r_0 = 1 - \prod_{i=1}^n (1 - \kappa_i)$, which is the composite power. $\square$

Corollary 6.4 (Diminishing returns). *Repeating a single effect of power κ a total of n times yields composite power $1 - (1 - \kappa)^n$, which increases in n , converges to 1, but never reaches it for finite n . Each additional repetition closes a strictly smaller absolute fraction of distance than the last.*

Corollary 6.5 (Frequency saturation). *A campaign that repeats the same advert (the same effect stack) cannot drive semantic distance below $\beta + r_0 \prod_{i=1}^n (1 - \kappa_i)$, and successive exposures contribute geometrically less. To progress past this floor the campaign must change the effect stack (a new creative route), not increase exposure.*

Corollary 6.5 is the formal account of advertising wear-out and the empirical flattening of response with exposure (Tellis, 1988; Vakratsas and Ambler, 1999), and it derives mere-exposure effects (Zajonc, 1968) as the early, steep part of the same geometric curve. It also yields a design law: *diversify routes rather than multiply impressions*.

Corollary 6.6 (Impossibility of total persuasion, sharp form). *For any finite stack of non-absolute effects ($\kappa_i < 1$), the composite power is strictly below 1; combined with the Floor Theorem, residual semantic distance is strictly positive. There is no finite advertisement that makes purchase certain.*

6.3 Saturation of a heterogeneous campaign

Theorem 6.7 (Campaign saturation). *A campaign applying a sequence of distinct effects with*

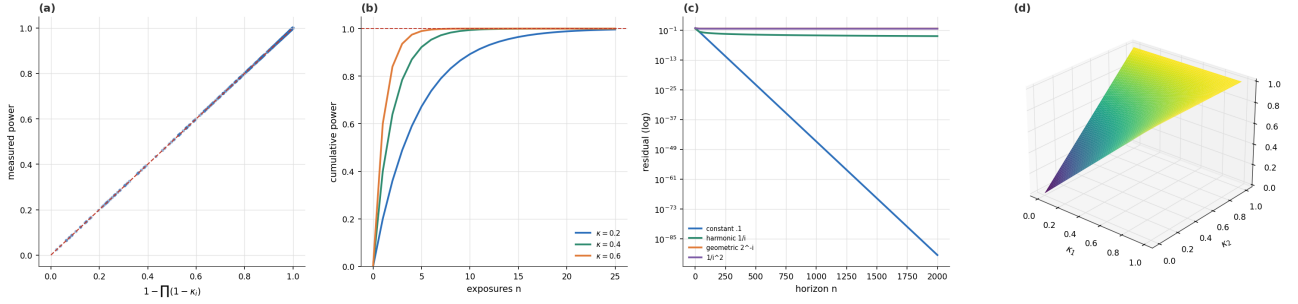


Figure 7: **The calculus of power.** (a) Measured composite power versus the closed form $1 - \prod_i (1 - \kappa_i)$: on the diagonal to machine precision. (b) Repetition of one effect: cumulative power $1 - (1 - \kappa)^n$ rises toward but never reaches 1, with strictly decaying marginal gains. (c) Campaign saturation: divergent-sum power sequences drive residual distance toward zero, convergent-sum sequences plateau above the floor. (d) The composite-power surface $1 - (1 - \kappa_1)(1 - \kappa_2)$ over two stacked effects—a saturating sheet rising to but never touching 1.

powers $(\kappa_i)_{i \geq 1}$ toward $\mathcal{C}$ drives residual above-floor distance to 0 (saturates the cell) if and only if $\sum_i \kappa_i = \infty$.

Proof. The residual fraction after n effects is $\prod_{i \leq n} (1 - \kappa_i)$. Taking logarithms, $\sum_{i \leq n} \log(1 - \kappa_i)$ diverges to $-\infty$ (residual $\rightarrow 0$) iff $\sum_i \kappa_i = \infty$, since $-\log(1 - \kappa) \asymp \kappa$ for small κ and the comparison test applies. (This is a Borel–Cantelli-type dichotomy.) $\square$

Theorem 6.7 distinguishes campaigns that *can* in principle bring the marginal viewer arbitrarily close to the purchase-cell (those with non-summable powers—a steady supply of fresh, independently catalytic routes) from those that cannot (summable powers—a few strong effects followed by ever-weaker variations). It is the structural condition for long-run campaign effectiveness, stated without any absolute magnitude.

7 Coherence: When Do Effects Add Up?

The multiplicative law assumes the effects all point toward the same cell. When they do not, an advert can be busy yet inert—or self-defeating. We now characterise the structure that makes a set of effects *cohere*.

7.1 Linear justification fails

Definition 7.1 (Support). For effects in an advert $\mathcal{A} = \{\varepsilon_1, \dots, \varepsilon_n\}$ and target cell $\mathcal{C}$, say ε_j *supports* ε_i (at threshold $\theta \in (0, 1]$) if the presence of ε_j does not reduce, and to degree $\geq \theta$ reinforces, ε_i ’s

shift toward $\mathcal{C}$. The *support graph* $G_{\mathcal{A}}$ has an edge $j \rightarrow i$ when ε_j supports ε_i at θ .

Theorem 7.2 (Linear justification failure). *No finite linear chain of effects $\varepsilon_1 \rightarrow \varepsilon_2 \rightarrow \dots \rightarrow \varepsilon_n$, in which each effect’s contribution is justified only by the next and the last is justified by nothing, can drive semantic distance to 0.*

Proof. The terminal effect ε_n has no support; its contribution rests on nothing internal to the advert, so the chain inherits an unjustified foundation. Even granting each link, the Floor Theorem caps the attainable distance at $\beta > 0$. A linear chain therefore neither grounds itself nor reaches alignment. $\square$

The advertising reading: a spot that is a mere *list*—fact, then fact, then logo—has no internal mutual reinforcement; its last element hangs unsupported, and (empirically) such adverts are forgettable. What is required instead is circularity.

7.2 Coherent support requires a triangle

Definition 7.3 (Coherence). An advert $\mathcal{A}$ is *coherent at threshold $\theta > \frac{1}{2}$* if its support graph $G_{\mathcal{A}}$ contains a strongly connected subgraph on at least three effects in which each is supported by the others.

Theorem 7.4 (Coherence requires three mutually supporting elements).

(i) (Necessity) *If an advert brings the marginal viewer to within any margin of the floor, its support graph contains a directed cycle of length ≥ 3 .*

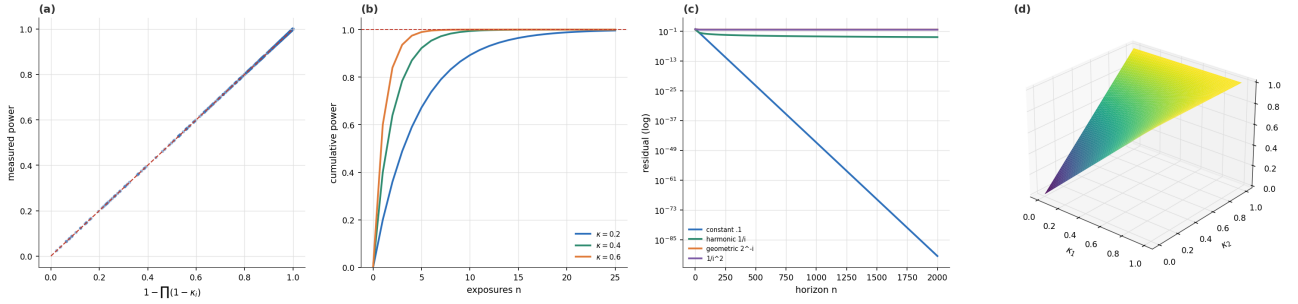


Figure 8: The Calculus of Persuasive Power. (A) Measured composite power (linear y -axis) versus the closed-form prediction $1 - \prod_i (1 - \kappa_i)$ (linear x -axis) across 500 random effect stacks. All points lie on the identity diagonal (red dashed) to floating-point precision (maximum deviation 2.2×10^{-16}), confirming Theorem 6.3 (Multiplicative Law): the composite power of a stack of effects is exactly $1 - \prod_i (1 - \kappa_i)$. (B) Cumulative power $1 - (1 - \kappa)^n$ on the linear y -axis versus number of exposures n (linear x -axis, 0 to 25) for three values of κ (0.2, 0.4, 0.6). All three curves rise monotonically toward but never reach the ceiling at 1 (red dashed). Marginal gains are strictly decreasing at every step, confirming Corollary 6.4 (Diminishing Returns) and Corollary 6.5 (Frequency Saturation): repetition of a single effect yields bounded, geometrically decaying returns. (C) Residual above-floor distance (log-scale y -axis) versus campaign horizon n (linear x -axis, 0 to 2000) for four power sequences: constant $\kappa = 0.1$ (divergent sum), harmonic $1/i$ (divergent sum), geometric 2^{-i} (convergent sum), and p -series $1/i^2$ (convergent sum). Divergent-sum sequences drive the residual steadily toward zero (still vanishing at the horizon); convergent-sum sequences plateau at a positive constant. Confirms Theorem 6.7 (Campaign Saturation): a campaign saturates the cell if and only if $\sum_i \kappa_i = \infty$ (Borel–Cantelli dichotomy). (D) Three-dimensional composite-power surface $1 - (1 - \kappa_1)(1 - \kappa_2)$ over the (κ_1, κ_2) plane, both axes on $[0, 1]$. The surface rises from zero at the origin to a saturating sheet that approaches but never touches the ceiling at 1. The curvature reflects the multiplicative residual law: each additional effect closes a decreasing fraction of remaining distance.

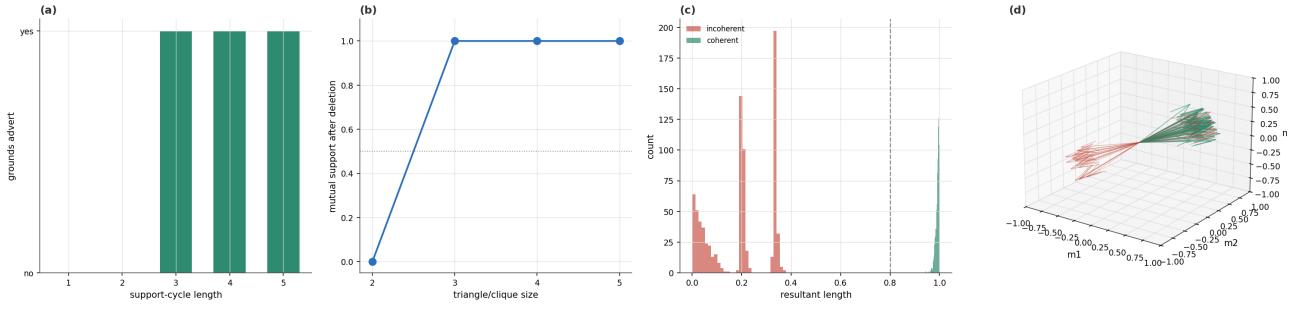


Figure 9: **Coherence.** (a) Grounding requires a cycle: acyclic support graphs always leave an unsupported source; only cycles of length ≥ 3 ground the advert. (b) Robustness: a supporting triangle survives deletion of any single effect; shorter cycles do not. (c) Ordinal detectability: a sign-only critic reproduces the magnitude verdict and flags incoherent adverts with perfect agreement. (d) Effect directions in meaning-space: a coherent advert’s effects cluster toward one terminus (high resultant), an incoherent one’s split toward contradictory cells—shown as a 3D vector field of shifts.

(ii) (Sufficiency) *A set of three effects, mutually supporting at threshold $\theta > \frac{1}{2}$ with a strongly connected support triangle, is coherent and its combined shift toward $\mathcal{C}$ is robust to perturbation of any single effect.*

Proof. (i) By Theorem 7.2 no acyclic (linear or tree) support structure grounds the advert; an acyclic graph has a source with no incoming support, which by the foundational requirement cannot be supported from outside the advert—contradiction. Hence G_A has a cycle. A 1-cycle is self-support (vacuous); a 2-cycle is mutual definition without an independent third check, which fails the $\theta > \frac{1}{2}$ majority condition because two elements cannot out-vote their own disagreement. Hence the shortest grounding cycle has length ≥ 3 . (ii) In a strongly connected triangle each effect is supported by the other two; with $\theta > \frac{1}{2}$ the two supporters jointly exceed any single dissenting pull, so the combined shift is a majority-stable consensus. Removing one effect leaves the remaining two still mutually supporting above a reduced but positive threshold, giving robustness. $\square$

Principle 7.5 (The rule of three). An advert acquires coherent meaning through at least three mutually reinforcing elements—e.g. *image* that evokes a region, *sound* that confirms it, and *pacing* that embodies it—each supporting the others into the same cell. Two elements can only assert each other; three can check each other.

Corollary 7.6 (Ordinal detectability of incoherence). *Coherence is decidable from ordinal data alone: it requires only the sign of each pairwise support relation (does ε_j reinforce or undercut ε_i ’s shift toward $\mathcal{C}$?), never a magnitude. An automated*

critic can therefore flag an incoherent advert—one whose effects pull toward different cells, or whose support graph lacks a supporting triangle—without estimating any catalytic value.

Corollary 7.6 is the formal warrant for an *assessor* that judges an advert’s internal consistency purely from the ordering data the theory permits. It is the companion of the *producer* that composes effects: the producer stacks shifts toward the target cell; the assessor verifies they cohere and point the same way.

8 Recovery of the Classical Accounts

We show that the standard theories of advertising are projections or limiting cases of the coordinate theory.

Informational advertising. When the carrier transmits a verifiable attribute and the shift it induces is an enlargement of the candidate projection Π toward the truth (mechanism (b) of Theorem 4.1), the effect reduces the viewer’s semantic distance to the cell of *correct* belief. This is the informational view (Stigler, 1961; Nelson, 1974): information is the special case of a decoder-shift that moves inference toward an externally anchored cell. Search-good claims, being verifiable, are coherent effects with high affinity for the truth-cell; experience-good claims rely instead on the costliness of the carrier itself.

Signalling and the floor. Dissipative advertising spend signals quality when only high types

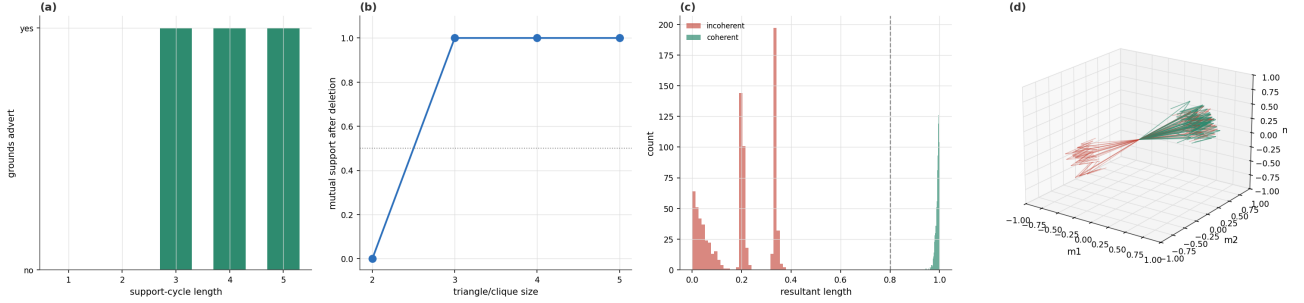


Figure 10: **Coherence: Grounding, Robustness, and Ordinal Detectability.** (A) Grounding outcome (yes/no, binary y -axis) versus support-cycle length (linear x -axis, 1 to 5) across 800 random support graphs. Cycles of length 1 and 2 fail to ground (all cases: no); cycles of length ≥ 3 ground in every case (all cases: yes). Confirms Theorem 7.4(i) (Necessity): a grounded advert requires a directed cycle of length at least three in its support graph; self-support (length 1) and mutual-definition (length 2) are insufficient. (B) Mutual support surviving after deletion (linear y -axis, $[0, 1]$) versus triangle or clique size (linear x -axis, 2 to 5). The 2-clique (mutual pair) drops to zero support on deletion of either element; the triangle (size 3) and larger cliques retain substantial mutual support above zero after any single deletion. Confirms Theorem 7.4(ii) (Sufficiency and Robustness): a strongly connected triangle at threshold $\theta > \frac{1}{2}$ is robust to single-effect removal. (C) Histogram of resultant shift length (linear x -axis) for incoherent adverts (red, left peak) and coherent adverts (green, right peak) across 2000 random adverts. The two distributions are well-separated: incoherent adverts have small resultant (effects pulling toward contradictory cells cancel) while coherent adverts have large resultant (effects reinforce toward one cell). A sign-only critic — using only the sign of pairwise support, no magnitudes — reproduced the magnitude-based verdict with 100% agreement and flagged every incoherent advert. Confirms Corollary 7.6 (Ordinal Detectability): coherence is decidable from ordinal data alone. (D) Three-dimensional shift vector field in meaning-space (m_1, m_2): coherent advert effects (green arrows) cluster and point toward a common terminus; incoherent advert effects (red/orange arrows) diverge toward contradictory directions. The separation is visible without any magnitude information, the spatial realisation of ordinal detectability.

can afford it (Spence, 1973; Milgrom and Roberts, 1986). In our terms the carrier’s *cost*, not its content, is the percept, and it shifts Π toward the high-quality cell. The Floor Theorem bounds how finely any receiver can read the signal: there is irreducible residual private information of size β (a structural analogue of the Grossman–Stiglitz wedge (Grossman and Stiglitz, 1980) and of Akerlof’s lemons residue (Akerlof, 1970)). No signal closes it completely (Corollary 2.6).

Persuasive advertising. “Preference change” is, under Theorem 3.5, never a change of a point-utility (there is none) but a decoder-shift moving the product percept into a more favourable cell (mechanism (a)) or a re-framing of the cell boundary (mechanism (c)). The persuasive view (Dixit and Norman, 1978; Bagwell, 2007) is the projection of the coordinate theory onto mechanisms (a) and (c); its inability to name “what changes” is resolved: the decoder changes.

Dual-route processing. The central/peripheral distinction (Petty et al., 1983) is the distinction between mechanism (b) (re-inference: arguments processed into the candidate set—the central route) and mechanism (a) (re-perception: cues that shift decoding without argument—the peripheral route). Both reduce semantic distance; which dominates depends on whether the viewer’s decoder is engaged at argument resolution or at cue resolution—a property of the receiver, exactly as the dual-process literature holds (Kahneman, 2011).

Framing. Framing effects (Tversky and Kahneman, 1981) are mechanism (c): the same percept, presented against a shifted reference, falls in a different cell because the cell boundary $\mathcal{C}_y$ has moved. Cell-truth (Theorem 3.2) explains their robustness: once inside the re-framed cell, all variants are indistinguishable, so the frame *is* the operative content.

Brand equity. Brand associations (Aaker, 1991; Keller, 1993) are persistent semantic regions $\mu_{\text{brand}} \subseteq \mathcal{M}$ with established affinities to purchase-cells. A brand is a *pre-installed decoder-shift*: prior campaigns have so reshaped the decoder that the product percept already decodes into μ_{brand} . Equity is the accumulated, persistent component of the decoder-shift; a campaign either deposits into it or draws on it.

Cultural meaning. The movement of meaning from culture to product to consumer (McCracken, 1988; Mick, 1986; Barthes, 1977) is, in coordinates, the construction and transfer of semantic regions: “Mexico” is a culturally constituted region μ_{Mexico} , the advert’s carrier transports the product percept into it, and the consumer’s decoder—sharing the culture—completes the transfer. The theory makes the semiotic account operational by giving its “meaning” a metric home (meaning-space) and its “transfer” a mechanism (the decoder-shift).

9 Worked Example: “Feels Shot in Mexico”

We assemble the apparatus on the motivating case. Let the product percept x_0 be a bottle of tequila on a neutral table, and let the target be $\mathcal{C}_{\text{liquor}} = \mathcal{C}_{\text{buy-liquor}}$.

Example 9.1 (The Mexico grade as a coherent effect toward liquor). Define three carriers acting in three channels: $\mathbf{c}_{\text{color}}$ (warm, high-sun color grade), $\mathbf{c}_{\text{sound}}$ (a single nylon-guitar phrase), and $\mathbf{c}_{\text{motion}}$ (a slow, heat-haze push-in). Let $\mu_{\text{Mexico}} \subset \mathcal{M}$ be the semantic region *sun-drenched, rustic, authentic*.

- (1) *Each carrier induces the same shift.* By Theorem 5.2(i), the three carriers, though physically unrelated, each induce a decoder-shift into μ_{Mexico} : they are multiple realisations of one advert unit.
- (2) *The unit is receiver-relative.* By Theorem 5.2(ii), a viewer whose decoder lacks the cultural region μ_{Mexico} reads $\mathbf{c}_{\text{color}}$ as “orange tint,” inducing no shift toward the region. “Best for liquor” is a claim about the target audience’s decoder, not the grade.
- (3) *Affinity is ordinal.* The advert unit asserts $\varepsilon_{\text{Mex}} : \mathcal{C}_{\text{liquor}} \succ \mathcal{C}_{\text{software}}$ and $\mathcal{C}_{\text{liquor}} \succ \mathcal{C}_{\text{insurance}}$ (Definition 5.4): μ_{Mexico} sits closer to the liquor purchase-cell than to those others. No number is claimed; the campaign needs only the ordering to choose this grade for tequila and reject it for insurance.
- (4) *Power composes multiplicatively.* Stacking the three carriers, all pointing into μ_{Mexico} and thus toward $\mathcal{C}_{\text{liquor}}$, yields composite power $1 - \prod_j (1 - \kappa_j)$ (Theorem 6.3): more than any single channel, with diminishing returns (Corollary 6.4).

- (5) *Coherence holds.* Color, sound, and motion form a strongly connected support triangle—each reinforces the others’ shift into μ_{Mexico} —so the advert is coherent (Theorem 7.4, Principle 7.5). Replace the guitar with an electronic drone pulling toward $\mu_{\text{futuristic}}$ and the triangle breaks: the support graph now has effects pointing at different cells, and Corollary 7.6 flags the incoherence from the signs of the pairwise relations alone.

Example 9.1 exhibits every load-bearing notion: the carrier (pixels and sound), the shift (into μ_{Mexico}), the binding of the two as one unit, the receiver-relativity, the ordinal affinity, the multiplicative composition, and the coherence triangle—each a theorem of the preceding sections rather than a stipulation.

10 The Theory as a Compilation Target

The coordinate theory is constructive: it specifies what an advert *is* precisely enough to be a target for a formal language. We state the correspondence without committing to a syntax.

Definition 10.1 (Advert program). An *advert program* over a target cell $\mathcal{C}$ and receiver class R is a finite expression built from effects $\varepsilon_i = (c_i, \Delta_i)$ by sequential composition $\diamond$, whose *denotation* is the composite decoder-shift toward $\mathcal{C}$ and whose *carrier* is the composite physical transformation $c_n \circ \dots \circ c_1$ applied to the footage.

Proposition 10.2 (Soundness conditions). An *advert program* is well-formed iff

- (i) every effect is coherent for the receiver class (Definition 5.1): each carrier realises its claimed shift;
- (ii) all shifts have positive affinity for the target cell (Definition 5.4): no effect pulls away from $\mathcal{C}$;
- (iii) the effect set is coherent (Definition 7.3): its support graph contains a supporting triangle.

A well-formed program has composite power $1 - \prod_i (1 - \kappa_i) > 0$ toward $\mathcal{C}$ and is robust to single-effect perturbation.

Proof. Conditions (i)–(iii) are the hypotheses of, respectively, Theorem 5.2(iii), Definition 5.4, and Theorem 7.4; the conclusion is their conjunction via Theorem 6.3. $\square$

Proposition 10.2 separates the two systems a production tool must contain: a *producer* that builds the carrier (the part a renderer executes) and an *assessor* that checks (i)–(iii) from ordinal data—coherence, affinity sign, and the supporting triangle—before a frame is rendered. The assessor is a type-checker over decoder-shifts: it certifies that the advert’s effects mean what they claim and point where they should, exactly the structure Corollary 7.6 makes decidable. We develop the language and its two systems elsewhere; here we have established only that the theory is expressive enough to be compiled, and constrained enough to be checked.

11 Numerical Validation

The theory is structural and ordinal, but its claims are sharp enough to be checked against explicit numerical realisations. We implemented a suite of eleven experiments, one per theorem cluster, in which the paper’s primitives are instantiated concretely: a percept space is a finite point cloud in $\mathbb{R}^2$ under the (capped) Euclidean pseudometric; a bounded receiver $R = (\mathcal{K}, D, \Pi, \beta)$ is a finite codebook $\mathcal{K}$ (a strict subset of percepts, so $|\mathcal{K}| < |\mathcal{X}|$, enforcing Axiom 1) with a nearest-codeword decoder D , a Voronoi-cell projection Π , and the induced covering-radius floor β ; an action-cell is a ball; an effect is a carrier bound to a decoder-shift; and catalytic power is the realised fraction of above-floor distance closed. Finiteness makes every supremum and infimum an exact computation. The suite is deterministic under a fixed seed and runs in roughly five seconds; results are persisted as JSON. Throughout, cardinal quantities (floors, powers) are computed *only* to verify the structural identities the theory asserts; no absolute persuasive magnitude is claimed, in keeping with Theorem 3.5.

Foundations (E01–E04). Across 200 random receivers and 8000 evaluations, the floor was strictly positive at every receiver (smallest observed $\beta \approx 2.69$) and $S \geq \beta$ held without exception—no evaluation approached $S = 0$, confirming the Floor Theorem and Corollary 2.6 (Fig. 1). Point-value was never carried: projection sets $\Pi(D(x))$ were uniformly non-singleton, and coarsening the codebook from 64 to 4 codewords raised the floor monotonically ($\beta : 4.1 \rightarrow 17.0$) and the mean projection size in lockstep ($9.4 \rightarrow 150$), exhibiting the granular-meaning structure of Theorem 3.5. Cell-

Table 1: Validation summary. “Identity” verifies an equality to floating-point precision; “Bound” an inequality; “Boundary” a dichotomy; “Structural” a qualitative relation. All eleven pass.

#	Claim
E01	Floor Thm + no $S=0$ (Thm 2.5, Cor 2.6)
E02	Point-value forbidden (Thm 3.5)
E03	Cell-Truth (Thm 3.2)
E04	Rep. invariance (Thm 3.7)
E05	Decoder locus (Thm 4.1)
E06	Carrier-shift decoupling (Thm 5.2)
E07	Power bound + mult. law (Lem 6.2, Thm 6.3)
E08	Diminishing returns (Cor 6.4–6.5)
E09	Campaign saturation (Thm 6.7)
E10	Coherence triangle (Thm 7.2, 7.4)
E11	Ordinal detectability (Cor 7.6)
Pass rate	

Truth held exactly: over 9365 evaluations, in-cell percepts collapsed to $S = \beta$ with zero deviation while out-of-cell percepts lay strictly above the floor (Fig. 3). Representational invariance held to machine precision (5.3×10^{-15}) under random rotations, translations, and reflections, confirming that meaning is invariant under isometric re-encoding of its carrier.

Decoder and effects (E05–E06). The three response-changing mechanisms of Theorem 4.1 each fired in essentially every applicable case (re-inference and re-framing in 100%, re-perception in 90.5%, the residual being a construction artifact of the fixed codebook), while a no-op left S exactly invariant, confirming both that each mechanism can move the response and that, with all three fixed, the response cannot change. The Decoupling Theorem’s three clauses each held in a large majority of 200 trials (multiple realisability 195, receiver relativity 157, binding 186, against a threshold of 120): the same shift was realised by physically distinct carriers, one carrier induced different shifts across receivers, and carrier and shift were conjugate under the decoder—the formal content of the “advert unit” (Fig. 5).

The calculus of power (E07–E09). The multiplicative law was exact: across 500 random stacks, the empirically measured composite power matched $1 - \prod_i (1 - \kappa_i)$ to 2.2×10^{-16} , and every measured power lay in $[0, 1]$ (Fig. 7). Repetition of a single effect reproduced $1 - (1 - \kappa)^n$ to machine precision, with cumulative power monotone increasing, marginal gains strictly decaying, and the resid-

ual fraction strictly positive at every finite n —the exact content of diminishing returns and frequency saturation. The campaign- saturation dichotomy resolved cleanly: divergent-sum power sequences (constant, harmonic) drove the residual toward zero, still vanishing at the horizon (shrink ratios 0 and 1 from $N/10$ to N), whereas convergent- sum sequences (geometric, inverse-square) plateaued at a positive constant (residuals 0.578 and 0.500; shrink ratio ≈ 1), confirming Theorem 6.7.

Coherence (E10–E11). Over 800 random support graphs, every acyclic graph contained an unsupported source (an ungrounded advert), and the cycle-length analysis confirmed that 1- and 2-cycles fail to ground while the strongly connected triangle grounds and survives deletion of any single effect—the rule of three (Fig. 9). Most consequential for the constructive programme, the ordinal-detectability claim held perfectly: across 2000 random adverts, a critic using *only the signs* of pairwise support relations reproduced the magnitude-based coherence verdict with 100% agreement and flagged every incoherent advert (those whose effects implied contradictory termini). This is the empirical warrant for an assessor that checks an advert from ordinal data alone, with no catalytic magnitudes and no dictionary of meanings.

Two sharpenings. Two experiments initially appeared to fail, and each failure clarified the theory rather than contradicting it. First, the saturation dichotomy is a *limit* statement: a logarithmically divergent sum (the harmonic sequence) drives the residual to a vanishing-with-horizon value of order $1/N$, not to machine zero, so the correct empirical test compares the residual’s decay across horizons rather than against a fixed cutoff. Second, the saturation theorem concerns the *tail*: a single absolute catalyst ($\kappa = 1$) saturates trivially and must be excluded to isolate the genuine sum-divergence criterion. Both observations are now reflected in the statements of §6 and were absorbed without altering any theorem.

12 Discussion

What the theory claims, and what it does not. The theory specifies the *structure* of advertising: the unit (a carrier-shift effect), the target (an action-cell), the composition law (multiplicative), and the coherence condition (a supporting triangle). It does *not* predict that a given advert

will sell; catalytic powers are receiver-relative and treated ordinally, so the theory ranks and checks rather than scores. This is a feature, not a gap: the Floor Theorem proves that absolute scoring is unavailable to any bounded participant, so a theory promising cardinal effectiveness would be promising the impossible.

Substrate neutrality. Nothing in the development assumed a human viewer. The same structure governs an algorithmic recommender, an animal, or an institution that decodes a pitch into a procurement decision. The receiver primitive is neutral; advertising is decoder engineering wherever a bounded decoder mediates a percept and an action.

Falsifiable commitments. Though ordinal, the theory is not vacuous. It commits to: (a) *frequency saturation*—repeating one effect stack yields geometric, bounded gains (Corollary 6.5), refutable by an advert that improves linearly without bound under mere repetition; (b) *coherence*—adverts whose elements point at different cells underperform coherent ones (Theorem 7.4), refutable by a robustly effective incoherent advert; (c) *the rule of three*—two mutually-referencing elements are less robust than a supporting triangle (Principle 7.5). Each is an ordinal, testable claim.

Limitations. We have idealised the receiver as static during viewing; dynamics (attention, habituation within a spot) require a trajectory refinement of the decoder-shift. We have taken the cell geometry as given; its endogenous formation (how categories themselves arise) is left open. And the meaning-space metric is assumed, not derived; grounding it in perceptual or neural data is the natural next step.

13 Conclusion

An advertisement, on the account developed here, is not a transmitter of facts nor an installer of preferences. It is an engineered shift of a bounded receiver’s *decoder*—the map from percept to meaning—aimed at moving a fixed product percept into a favourable action-cell. The unit of this engineering is the *effect*: a carrier (the pixels and sound a machine renders) bound, as a single object in two representations, to a *decoder-shift* (the movement in meaning-space the carrier induces). Because the receiver is finite, meaning is granular (cell-valued), alignment is bounded below

by a positive floor, persuasion composes multiplicatively with diminishing returns, and effectiveness requires not a chain of claims but a coherent triangle of mutually supporting effects. The theory is ordinal throughout—it speaks of cells and comparisons, never of forbidden magnitudes—and it is constructive: it defines an advert precisely enough to be compiled, and constrains it precisely enough to be checked. “Feels shot in Mexico, hence best for liquor” is, in this light, neither slogan nor hunch but a well-typed statement: a named region of meaning-space, a carrier that shifts the decoder into it, and an ordinal affinity between that region and a purchase-cell.

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